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1 Professional Preparation

1. Postdoctoral research associate. University of Illinois at Urbana-Champaign. 1998-9. Adviser: Dr. Daniel Reed.
2. Ph.D. in Computer Science. Georgia Institute of Technology. Atlanta. 1998. Adviser: Dr. Karsten Schwan.
3. M.S. in Computer Science. Georgia Institute of Technology. Atlanta. 1994.
4. B.S. in Electrical/Computer Engineering with Honors. Georgia Institute of Technology. Atlanta. 1989.

2 Appointments

1. Oak Ridge National Laboratory.
   (a) Distinguished R&D Staff Member and Group Leader. November 2009 - present.
   (b) Founding Group Leader of ORNL Future Technologies Group. 2004 - present.
   (c) Founding Director of Experimental Computing Laboratory (ExCL). 2004 - present.
   (d) Senior R&D Staff Member. November 2003 - November 2009
2. University of Tennessee - Knoxville, Department of Electrical Engineering and Computer Science.
   (a) Joint Professor. February 2016 - present
3. Georgia Institute of Technology, College of Computing.
   (a) Joint Professor. October 2006 - 2016.
   (b) Principal Investigator and Director, NVIDIA CUDA Center of Excellence. 2010 - 2016.
   (c) Principal Investigator and Project Director, NSF Keeneland Project. 2009 - 2015.
   (d) Adjunct Professor. May 2005 - October 2006.
4. Lawrence Livermore National Laboratory. Computer Scientist and Project Leader. 1999 - 2003.
5. Los Alamos National Laboratory. Graduate research intern. Summers of 1996, 1997.
6. Intel Corporation. Graduate research intern. Sept 1991 - Dec 1992, Summer 1993.
7. Hewlett Packard Corporation (now Agilent Technologies). Technical staff member. 1989 - 1991.
8. Georgia Institute of Technology, Undergraduate research assistant. Experiment control system for analysis of CCD semiconductor materials. University semesters during 1987-1989.

3 Honors

1. Institute of Electrical and Electronic Engineers
   - **Senior** member, 2012.
   - **Fellow**, 2016.
2. ACM
   - **Distinguished Scientist** member, 2012.
3. **ACM/IEEE Gordon Bell Prize**: "Petascale Direct Numerical Simulation of Blood Flow on 200K Cores and Heterogeneous Architectures." A. Rahimian, I. Lashuk, S. Veerapaneni, A. Chandramowlishwaran, D. Malhotra, L. Moon, R. Sampath, A. Shringarpure, J. Vetter, R. Vuduc, D. Zorin and G. Biros (2010). 2010 ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC10).
4. **Best Paper**: "NVIDIA Tensor Core Programmability, Performance & Precision," S. Markidis, S. Wei Der Chien, E. Laure, I.B. Peng, and J.S. Vetter, Proc. Eighth International Workshop on Accelerators and Hybrid Exascale Systems (AsHES) 2018.
5. **Best Student Paper Finalist**: L. Yu, D. Li, S. Mittal, J.S. Vetter, "Quantitatively Modeling Application Resiliency with the Data Vulnerability Factor," in SC14: ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis. New Orleans, LA, 2014
6. **Best Paper Finalist**: M.E. Belviranli, S. Lee, and J.S. Vetter, "Designing Algorithms for the EMU Migrating-threads-based Architecture," in IEEE HPEC, 2018.
7. **Award Paper**: S.R. Alam and J.S. Vetter, Hierarchical Model Validation of Symbolic Performance Models of Scientific Applications, Proc. Euro-Par 2006.
8. **Award Paper**: Vetter and F. Mueller (2002). Communication Characteristics of Large-Scale Scientific Applications for Contemporary Cluster Architectures. International Parallel and Distributed Processing Symp. (IPDPS), 2002.
9. **Best Presentation Award**: Exascale Applications and Software Conference (EASC) for "Exploring Emerging Technologies in the Extreme Scale HPC Co-Design Space with Holistic Performance Modeling," 2015.
10. **Most Distinguished Software**: ORNL Computer Science and Mathematics Division Award for Scalable Heterogeneous Computing Benchmark suite, 2013.
11. NASA Graduate Research Fellowship, 1993-7.
12. LLNL Computation Directorate Award for ASCI PSE milestone contributions, 2001.
13. LLNL Computation Directorate Award for technical contributions on ASCI Purple and BlueGene/L procurement, 2002.
14. LLNL Computation Directorate Award for ASCI performance modeling and analysis contributions, 2003.

4 Publications

Google Scholar citations are available at `http://scholar.google.com/citations?hl=en&user=pWnx6r4AAAAJ`.

4.1 Books

1. J. S. Vetter, editor. *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 2 of *CRC Computational Science Series*. Taylor and Francis, Boca Raton, 1 edition, 2015a

2. J. S. Vetter, editor. *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 1 of *CRC Computational Science Series*. Taylor and Francis, Boca Raton, 1 edition, 2013a

4.2 Book Chapter

1. J. S. Vetter. Terascale to petascale: The past 17 years in high performance computing. In *Contemporary High Performance Computing*, volume 2 of *Chapman and Hall/CRC Computational Science*, pages 1–5. Taylor and Francis, Boca Raton, 1 edition, 2015b. doi:doi:10.1201/b18256-2

2. J. S. Vetter, S. Lee, D. Li, G. Marin, C. McCurdy, J. Meredith, P. C. Roth, and K. Spafford. Quantifying architectural requirements of contemporary extreme-scale scientific applications. In Stephen A. Jarvis, Steven A. Wright, and Simon D. Hammond, editors, *High Performance Computing Systems. Performance Modeling, Benchmarking and Simulation: 4th International Workshop, PMBS 2013, Denver, CO, USA, November 18, 2013. Revised Selected Papers*, pages 3–24. Springer International Publishing, Cham, 2014. doi:10.1007/978-3-319-10214-6_1

3. J. S. Vetter, R. Glassbrook, K. Schwan, S. Yalamanchili, M. Horton, A. Gavrilovska, M. Slawinska, J. Dongarra, J. Meredith, P. C. Roth, K. Spafford, S. Tomov, and J. Wynkoop. Keeneland: Computational science using heterogeneous GPU computing. In Jeffrey S. Vetter, editor, *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 1 of *CRC Computational Science Series*, page 730. Taylor and Francis, Boca Raton, 1 edition, 2013a

4. J. S. Vetter. Contemporary high performance computing: an introduction. In Jeffrey S. Vetter, editor, *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 1 of *CRC Computational Science Series*, page 730. Taylor and Francis, Boca Raton, 1 edition, 2013b

5. J. S. Vetter, V. Tipparaju, W. Yu, and P. C. Roth. HPC interconnection networks: The key to exascale computing. In *Advances in Parallel Computing: High Speed and Large Scale Scientific Computing*, volume 18, pages 95–106. 2009

6. J. Dongarra, R. Graybill, W. Harrod, R. Lucas, E. Lusk, P. Luszczek, J. McMahon, A. Snavely, J. Vetter, K. Yelick, S. Alam, R. Campbell, L. Carrington, T.-Y. Chen, O. Khalili, J. Meredith, and M. Tikir. DARPA's HPCS program: History, models, tools, languages. In V. Zelkowitz Marvin, editor, *Advances in Computers*, volume Volume 72, pages 1–100. Elsevier, 2008. doi:10.1016/s0065-2458(08)00001-6

7. L. Derose, M. Pantano, J. S. Vetter, and D. A. Reed. Performance issues in parallel processing systems. In Gunter Haring, Christoph Lindemann, and Martin Reiser, editors, *Performance Evaluation - Origins and Directions*, page 527. Springer-Verlag, 2000

8. G. Eisenhauer, W. Gu, T. Kindler, K. Schwan, D. Silva, and J. S. Vetter. Opportunities and tools for highly interactive distributed and parallel computing. In Margaret Simmons, Ann Hayes, Jeffrey Brown, and Dan Reed, editors, *Debugging and Performance Tuning for Parallel Computing Systems*, pages 245–77. IEEE Computer Society Press, 1996

4.3 Journal Article

1. J. S. Vetter, E. P. DeBenedictis, and T. M. Conte. Architectures for the post-moore era. *IEEE Micro*, 37(4):6–8, 2017a

2. M. Umar, S. V. Moore, J. S. Meredith, J. S. Vetter, and K. W. Cameron. Aspen-based performance and energy modeling frameworks. *Journal of Parallel and Distributed Computing*, 2017. doi:https://doi.org/10.1016/j.jpdc.20

3. S. Mittal, R. Wang, and J. S. Vetter. DESTINY: A comprehensive tool with 3D and multi-level cell memory modeling capability. *Journal of Low Power Electronics and Applications*, 7(3):23–35, 2017a

4. E. Deelman, C. Carothers, A. Mandal, B. Tierney, J. S. Vetter, I. Baldin, C. Castillo, G. Juve, D. Krol, V. Lynch, B. Mayer, J. Meredith, T. Proffen, P. Ruth, and R. F. da Silva. PANORAMA: An approach to performance modeling and diagnosis of extreme-scale workflows. *International Journal of High Performance Computing Applications*, 31(1):4–18, 2017a. doi:10.1177/1094342015594515

5. E. Deelman, T. Peterka, I. Altintas, C. D. Carothers, K. K.v. Dam, K. Moreland, M. Parashar, L. Ramakrishnan, M. Taufer, and J. Vetter. The future of scientific workflows. *The International Journal of High Performance Computing Applications*, 0(0):1094342017704893, 2017b. doi:10.1177/1094342017704893

6. S. Mittal and J. Vetter. A technique for improving lifetime of non-volatile caches using write-minimization. *Journal of Low Power Electronics and Applications*, 6(1):1, 2016a

7. S. Mittal and J. S. Vetter. A survey of techniques for architecting DRAM caches. *IEEE Transactions on Parallel and Distributed Systems*, 27(6):1852–1863, 2016b. doi:10.1109/Tpds.2015.2461155

8. S. Mittal and J. S. Vetter. A survey of architectural approaches for data compression in cache and main memory systems. *IEEE Transactions on Parallel and Distributed Systems*, 27(5):1524–1536, 2016c. doi:10.1109/Tpds.2015.2435788

9. S. Mittal and J. S. Vetter. A survey of techniques for modeling and improving reliability of computing systems. *IEEE Transactions on Parallel and Distributed Systems*, 27(4):1226–1238, 2016d. doi:10.1109/Tpds.2015.2426179

10. S. Mittal and J. S. Vetter. A survey of software techniques for using non-volatile memories for storage and main memory systems. *IEEE Transactions on Parallel and Distributed Systems*, 27(5):1537–1550, 2016e. doi:10.1109/TPDS.2015.2442980

11. S. Mittal and J. S. Vetter. Equalwrites: Reducing intra-set write variations for enhancing lifetime of non-volatile caches. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 24(1): 103–114, 2016f. doi:10.1109/Tvlsi.2015.2389113

12. S. Mittal, J. S. Vetter, and L. Jiang. Addressing read-disturbance issue in STT-RAM by data compression and selective duplication. *IEEE Computer Architecture Letters*, 2016

13. J. S. Vetter and S. Mittal. Opportunities for nonvolatile memory systems in extreme-scale high performance computing. *Computing in Science and Engineering*, 17(2):73–82, 2015. doi:10.1109/mcse.2015.4

14. K. L. Spafford and J. S. Vetter. Automated design space exploration with aspen. *Scientific Programming*, 2015:10, 2015. doi:10.1155/2015/157305

15. A. Sabne, P. Sakdhnagool, S. Lee, and J. S. Vetter. Understanding portability of a high-level programming model on contemporary heterogeneous architectures. *IEEE Micro*, 35(4):48–58, 2015. doi:10.1109/MM.2015.73

16. S. Mittal, J. S. Vetter, and D. Li. A survey of architectural approaches for managing embedded DRAM and non-volatile on-chip caches. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 26 (6):1524–1537, 2015. doi:10.1109/tpds.2014.2324563

17. S. Mittal and J. S. Vetter. A survey of methods for analyzing and improving GPU energy efficiency. *ACM Computing Surveys*, 47(2):19:23, 2015a. doi:10.1145/2636342

18. S. Mittal and J. S. Vetter. A survey of CPU-GPU heterogeneous computing techniques. *ACM Computing Surveys*, 47(4):1–35, 2015b. doi:10.1145/2788396

19. S. Mittal and J. Vetter. A survey of techniques for modeling and improving reliability of computing systems. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 27(4):1226–1238, 2015c. doi:10.1109/tpds.2015.2426179

20. S. Mittal and J. Vetter. A survey of architectural approaches for data compression in cache and main memory systems. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 27(5):1524–1536, 2015d. doi:10.1109/tpds.2015.2435788

21. S. Mittal and J. S. Vetter. Equalwrites: Reducing intra-set write variations for enhancing lifetime of non-volatile caches. *IEEE Transactions on VLSI Systems*, 24(1):103–114, 2015e. doi:10.1109/tvlsi.2015.2389113

22. S. Mittal and J. S. Vetter. AYUSH: A technique for extending lifetime of SRAM-NVM hybrid caches. *IEEE Computer Architecture Letters (CAL)*, 14(2):115–118, 2015f. doi:10.1109/lca.2014.2355193

23. E. Deelman, C. Carothers, A. Mandal, B. Tierney, J. S. Vetter, I. Baldin, C. Castillo, G. Juve, D. Krol, V. Lynch, B. Mayer, J. Meredith, T. Proffen, P. Ruth, and R. Ferreira da Silva. PANORAMA: An approach to performance modeling and diagnosis of extreme-scale workflows. *International Journal of High Performance Computing Applications*, 2015. doi:10.1177/1094342015594515

24. L. Yu, D. Li, S. Mittal, and J. S. Vetter. Quantitatively modeling application resiliency with the data vulnerability factor. *ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, pages 695–706, 2014a. doi:10.1109/SC.2014.62

25. Y. Tian, C. Xu, W. Yu, J. Vetter, S. Klasky, H. Liu, and S. Biaz. neCODEC: nearline data compression for scientific applications. *Cluster Computing*, pages 1–12, 2013. doi:10.1007/s10586-013-0265-8

26. K. Spafford, J. S. Vetter, T. Benson, and M. Parker. Modeling synthetic aperture radar computation with aspen. *International Journal of High Performance Computing Applications*, 27(3):255–262, 2013. doi:10.1177/1094342013488262

27. A. Danalis, P. Luszczek, G. Marin, J. S. Vetter, and J. Dongarra. Blackjackbench: Portable hardware characterization with automated results' analysis. *The Computer Journal*, 2013

28. W. Yu, X. Que, V. Tipparaju, and J. S. Vetter. HiCOO: Hierarchical cooperation for scalable communication in global address space programming models on cray XT systems. *Journal of Parallel and Distributed Computing*, 2012. doi:10.1016/j.jpdc.2012.01.022

29. V. Tipparaju, E. Apra, W. Yu, X. Que, and J. Vetter. Runtime techniques to enable a highly-scalable global address space model for petascale computing. *International Journal of Parallel Programming*, 40(6):1–23, 2012. doi:10.1007/s10766-012-0214-9

30. A. Danalis, P. Luszczek, J. Dongarra, G. Marin, and J. S. Vetter. Blackjackbench: Portable hardware characterization. *SIGMETRICS Performance Evaluation Review*, 40(2), 2012a

31. J. S. Vetter, R. Glassbrook, J. Dongarra, K. Schwan, B. Loftis, S. McNally, J. Meredith, J. Rogers, P. Roth, K. Spafford, and S. Yalamanchili. Keeneland: Bringing heterogeneous GPU computing to the computational science community. *IEEE Computing in Science and Engineering*, 13(5):90–95, 2011. doi:10.1109/MCSE.2011.83

32. J. S. Meredith, P. C. Roth, K. L. Spafford, and J. S. Vetter. Performance implications of nonuniform device topologies in scalable heterogeneous architectures. *IEEE Micro*, 31(5):66–75, 2011. doi:10.1109/MM.2011.79

33. J. Dongarra, P. Beckman, T. Moore, P. Aerts, G. Aloisio, J.-C. Andre, D. Barkai, J.-Y. Berthou, T. Boku, B. Braunschweig, F. Cappello, B. Chapman, C. Xuebin, A. Choudhary, S. Dosanjh, T. Dunning, S. Fiore, A. Geist, B. Gropp, R. Harrison, M. Hereld, M. Heroux, A. Hoisie, K. Hotta, J. Zhong, Y. Ishikawa, F. Johnson, S. Kale, R. Kenway, D. Keyes, B. Kramer, J. Labarta, A. Lichnewsky, T. Lippert, B. Lucas, B. Maccabe, S. Matsuoka, P. Messina, P. Michielse, B. Mohr, M. S. Mueller, W. E. Nagel, H. Nakashima, M. E. Papka, D. Reed, M. Sato, E. Seidel, J. Shalf, D. Skinner, M. Snir, T. Sterling, R. Stevens, F. Streitz, B. Sugar, S. Sumimoto, W. Tang, J. Taylor, R. Thakur, A. Trefethen, M. Valero, A. van der Steen, J. Vetter, P. Williams, R. Wisniewski, and K. Yelick. The international exascale software project roadmap. *International Journal of High Performance Computing Applications*, 25(1): 3–60, 2011. doi:10.1177/1094342010391989

34. K. F. Alvin, B. W. Barrett, R. B. Brightwell, S. S. Dosanjh, A. Geist, K. S. Hemmert, M. A. Heroux, D. Kothe, R. C. Murphy, J. Nichols, R. A. Oldfield, A. F. Rodrigues, and J. S. Vetter. On the path to exascale. *International Journal of Distributed Systems and Technologies (IJDST)*, 1(2):1–22, 2010

35. B. de Supinski, S. Alam, D. Bailey, L. Carrington, C. Daley, A. Dubey, T. Gamblin, D. Gunter, P. Hovland, H. Jagode, K. Karavanic, G. Marin, J. Mellor-Crummey, S. Moore, B. Norris, L. Oliker, C. Olschanowsky, P. Roth, M. Schulz, S. Shende, A. Snavely, W. Spear, M. Tikir, J. S. Vetter, P. Worley, and N. Wright. Modeling the office of science ten year facilities plan: The PERI architecture tiger team. *Journal of Physics: Conference Series (Proceedings of SciDAC 2009, San Diego, CA, July 14-18, 2009)*, 2009

36. J. S. Meredith, G. Alvarez, T. A. Maier, T. C. Schulthess, and J. S. Vetter. Accuracy and performance of graphics processors: A quantum monte carlo application case study. *Parallel Comput.*, 35(3):151–163, 2009. doi:10.1016/j.parco.2008.12.004

37. S. R. Alam, P. K. Agarwal, and J. S. Vetter. Performance characteristics of biomolecular simulations on high-end systems with multi-core processors. *PARALLEL COMPUTING*, 34(11):640–651, 2008a. doi:10.1016/j.parco.2008.05.003

38. S. R. Alam, R. F. Barrett, M. R. Fahey, J. A. Kuehn, O. E. Bronson Messer, R. T. Mills, P. C. Roth, J. S. Vetter, and P. H. Worley. An evaluation of the oak ridge national laboratory cray XT3. *International Journal of High Performance Computing Applications*, 22(1):52–80, 2008b. doi:10.1177/1094342007085019

39. B. Mohr, L. DeRose, and J. S. Vetter. A framework for performance analysis of co-array fortran. *Concurrency and Computation: Practice and Experience*, 19(17):2207–2218, 2007

40. E. Grobelny, D. Bueno, I. Troxel, A. D. George, and J. S. Vetter. FASE: A framework for scalable performance prediction of HPC systems and applications. *SIMULATION*, 83(10):721–745, 2007. doi:10.1177/0037549707084939

41. D. H. Bailey, R. Lucas, P. Hovland, B. Norris, K. Yelick, D. Gunter, B.d. Supinski, D. Quinlan, P. Worley, J. Vetter, P. Roth, J. Mellor-Crummey, A. Snavely, J. Hollingsworth, D. Reed, R. Fowler, Y. Zhang, M. Hall, J. Chame, J. Dongarra, and S. Moore. Performance engineering: Understanding and improving the performance of large-scale codes. *Cyberinfrastructure Technology Watch Quarterly*, 3(4):55–62, 2007

42. S. R. Alam, P. K. Agarwal, M. C. Smith, J. S. Vetter, and D. Caliga. Using FPGA devices to accelerate biomolecular simulations. *IEEE Computer*, 40(2):66–73, 2007a

43. H. Ong, J. Vetter, R. S. Studham, C. McCurdy, B. Walker, and A. Cox. Kernel-level single system image for petascale computing. *ACM SIGOPS Operating Systems Review*, 40(2):50–54, 2006. doi:10.1145/1131322.1131335

44. R. Fatoohi, K. Kardys, S. Koshy, S. Sivaramakrishnan, and J. S. Vetter. Performance evaluation of high-speed interconnects using dense communication patterns. *Parallel Comput.*, 32(11-12):794–807, 2006. doi:10.1016/j.parco.2006.09.007

45. S. R. Alam, N. Bhatia, and J. S. Vetter. Symbolic performance modeling of HPCS applications. *Cyberinfrastructure Technology Watch*, 2(4B), 2006a

46. J. S. Vetter, B. R. de Supinski, J. May, L. Kissel, and S. Vaidya. Evaluating high performance computers. *Concurrency and Computation: Practice and Experience*, 17(10):1239–1270, 2005a. doi:10.1002/cpe.892

47. Jr. T. H. Dunigan, J. S. Vetter, III J. B. White, and P. H. Worley. Performance evaluation of the cray X1 distributed shared memory architecture. *IEEE Micro*, 25(1):30–40, 2005a

48. J. S. Vetter and F. Mueller. Communication characteristics of large-scale scientific applications for contemporary cluster architectures. *Journal of Parallel and Distributed Computing*, 63(9):853–865, 2003

49. J. S. Vetter and D. A. Reed. Real-time performance monitoring, adaptive control and interactive steering of computational grids. *International Journal of High Performance Computing Applications*, 14(4):357–366, 2000

50. J. S. Vetter and K. Schwan. Techniques for high-performance computational steering. *IEEE Concurrency*, 7(4):63–74, 1999

51. W. Gu, G. Eisenhauer, K. Schwan, and J. S. Vetter. Falcon: On-line monitoring and steering of parallel programs. *Concurrency: Practice and Experience*, 10(9):699–736, 1998

52. J. S. Vetter. Computational steering annotated bibliography. *SIGPLAN Notices*, 32(6):40–44, 1997

53. B. Schroeder, G. Eisenhauer, K. Schwan, J. Heiner, P. Highnam, V. Martin, and J. S. Vetter. From interactive applications to distributed laboratories. *IEEE Concurrency*, 6(2):78–90, 1997a

54. B. Schroeder, G. Eisenhauer, K. Schwan, J. Heiner, V. Martin, S. Szou, J. Vetter, R. Wang, F. Alyea, B. Ribarsky, and M. Trauner. Framework for collaborative steering of scientific applications. *Science Information Systems Newsletter*, IV(40):19–23, 1997b

55. W. Gu, J. S. Vetter, and K. Schwan. An annotated bibliography of interactive program steering. *SIGPLAN Notices*, 29(9):140–8, 1994

4.4 Peer-Reviewed Conference and Workshop Papers

1. M. E. Belviranli, S. Lee, J. S. Vetter, and L. N. Bhuyan. Juggler: a dependence-aware task-based execution framework for GPUs. In *Proceedings of the 23rd ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming*, pages 54–67, Vienna, Austria, 2018. ACM. doi:10.1145/3178487.3178492

2. S. Mittal, H. Wang, A. Jog, and J. S. Vetter. Design and analysis of soft-error resilience mechanisms for GPU register file. In *VLSI Design and 2017 16th International Conference on Embedded Systems (VLSID), 2017 30th International Conference on*, pages 409–414. IEEE, 2017b

3. S. Mittal, R. Bishnoi, F. Oboril, H. Wang, M. Tahoori, A. Jog, and J. S. Vetter. Architecting SOT-RAM based GPU register file. In *VLSI (ISVLSI), 2017 IEEE Computer Society Annual Symposium on*, pages 38–44. IEEE, 2017c

4. V. E. Lynch, J. B. Calvo, E. Deelman, R. F.d. Silva, M. Goswami, Y. Hui, E. Lingerfelt, and J. S. Vetter. Distributed workflows for modeling experimental data. In *2017 IEEE High Performance Extreme Computing Conference (HPEC)*, pages 1–5, 2017. doi:10.1109/HPEC.2017.8091071

5. J. Kim, S. Lee, and J. S. Vetter. PapyrusKV: a high-performance parallel key-value store for distributed NVM architectures. In *SC17: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–14, Denver, Colorado, 2017a. ACM. doi:10.1145/3126908.3126943

6. J. Kim, K. Sajjapongse, S. Lee, and J. S. Vetter. Design and implementation of papyrus: Parallel aggregate persistent storage. In *2017 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 1151–1162, Orlando, 2017b. IEEE. doi:10.1109/IPDPS.2017.72

7. J. E. Denny, S. Lee, and J. S. Vetter. Language-based optimizations for persistence on nonvolatile main memory systems. In *International Parallel and Distributed Systems Symposium (IPDPS)*, Orlando, 2017. IEEE

8. C. D. Carothers, J. S. Meredith, M. P. Blanco, J. S. Vetter, M. Mubarak, J. LaPre, and S. Moore. Durango: Scalable synthetic workload generation for extreme-scale application performance modeling and simulation. In *ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS)*, Singapore, 2017. ACM

9. P. Wu, D. Li, Z. Chen, J. S. Vetter, and S. Mittal. Algorithm-directed data placement in explicitly managed non-volatile memory. In *Proceedings of the 25th ACM International Symposium on High-Performance Parallel and Distributed Computing*, pages 141–152, Kyoto, Japan, 2016. ACM. doi:10.1145/2907294.2907321

10. J. S. Vetter. Preparing for supercomputing's sixth wave. In *Proceedings of the 25th ACM International Symposium on High-Performance Parallel and Distributed Computing*, pages 1–1. ACM, 2016

11. M. Umar, J. S. Meredith, J. S. Vetter, and K. W. Cameron. A study of power-performance modeling using a domain-specific language. In *2016 28th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)*, pages 84–92, 2016. doi:10.1109/SBAC-PAD.2016.19

12. S. Mittal and J. S. Vetter. Reducing soft-error vulnerability of caches using data compression. In *ACM Great Lakes Symposium on VLSI (GLSVLSI)*, 2016g

13. A. Mandal, P. Ruth, I. Baldin, D. Król, G. Juve, R. Mayani, R. F. Da Silva, E. Deelman, J. Meredith, and J. Vetter. Toward an end-to-end framework for modeling, monitoring and anomaly detection for scientific workflows. In *Parallel and Distributed Processing Symposium Workshops, 2016 IEEE International*, pages 1370–1379. IEEE, 2016

14. S. Lee, J. Kim, and J. S. Vetter. OpenACC to FPGA: A framework for directive-based high-performance reconfigurable computing. In *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, Chicago, 2016. IEEE. doi:10.1109/IPDPS.2016.28

15. J. Kim, S. Lee, and J. S. Vetter. IMPACC: A tightly integrated MPI+openACC framework exploiting shared memory parallelism. In *Proceedings of the 25th ACM International Symposium on High-Performance Parallel and Distributed Computing*, pages 189–201, Kyoto, Japan, 2016. ACM. doi:10.1145/2907294.2907302

16. J. E. Denny, S. Lee, and J. S. Vetter. NVL-C: Static analysis techniques for efficient, correct programming of non-volatile main memory systems. In *Proceedings of the 25th ACM International Symposium on High-Performance Parallel and Distributed Computing*, pages 125–136, Kyoto, Japan, 2016. ACM. doi:10.1145/2907294.2907303

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4.5 Other Reports

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2. J. Kim, S. Lee, and J. S. Vetter. An openACC-based unified programming model for multi-accelerator systems (poster). Technical report, ACM, 2015

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4. S. Mittal and J. S. Vetter. A survey of software techniques for using non-volatile memories for storage and main memory systems. Technical Report ORNL/TM-2014/633, Oak Ridge National Laboratory, 2014b

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11. J. S. Vetter, T. L. Windus, and B. Gorda. Performance metrics for high end computing. Prepared in response to the invitation to submit white papers to high end computing revitalization task force, 2003

12. F. Mueller, D. Lowenthal, J. S. Vetter, and V. Freeh. Performance analysis, scalability assessment and optimizations for high-end computing platforms. Prepared in response to the invitation to submit white papers to high end computing revitalization task force, 2003

13. C. F. McMillan, T. F. Adams, M. G. McCoy, R. B. Christensen, B. S. Pudliner, M. R. Zika, P. S. Brantley, J. S. Vetter, and J. M. May. Computational challenges in nuclear weapons simulation. Report UCRL-JC-155202, Lawrence Livermore National Laboratory, 2003

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15. S. Ashby, D. H. Bailey, M. Blackmon, P. Bohrer, K. Cameron, C. DeTar, J. Dongarra, D. Dwoyer, P. Freeman, A. Gheith, B. Gorda, G. Hammer, W. Felter, J. Kepner, D. Koester, S. McKee, D. Nelson, J. Nichols, M. Vahle, J. S. Vetter, T. Windus, and P. Worley. Performance modeling, metrics and specifications: Report of high end computing revitalization task force (HECRTF) working group six. Technical report, 2003

16. J. S. Vetter. Experiences using computational steering on existing scientific applications. Technical report, 1999

17. W. Ribarsky, J. Tumblin, G. Newton, R. Nowicki, and J. S. Vetter. GLYPHMAKER: An interactive, programmerless approach for customizing, exploring, and analyzing visual data representatives. Technical Report GIT-GVU-93-26, Graphics, Visualization, and Usability Center, Georgia Tech, 1994

4.6 PhD Thesis

1. J. S. Vetter. *Techniques and optimizations for high performance computational steering.* Ph.d., 1998

5 Presentations

5.1 Keynote

1. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , Exascale Application and Software Conference (EASC), Edinburgh, 2018.

2. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , 12th International Conference on Parallel Processing and Applications, Lublin, 2017.

3. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , ExaComm: Third International Workshop on Communication Architectures for HPC, Big Data, Deep Learning and Clouds at Extreme Scale, Frankfurt, 2017.

4. *Preparing for Supercomputing's Sixth Wave*, ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC), Kyoto, 2016.

5. *Exploring Emerging Technologies in the HPC Co-Design Space*, 11th International Conference on Parallel Processing and Applications, Krakow, 2015.

6. *Exploring Emerging Technologies in the HPC Co-design Space*, 2nd International HPC Forum, Tianjin, 2015.

7. *Exploring Emerging Technologies in the HPC Co-design Space*, International Conference on Energy-Aware High Performance Computing, Dresden, Germany, 2014.

8. *Exploring Emerging Technologies in the HPC Co-design Space*, AsHES Workshop (in conjunction with IPDPS), Phoenix, 2014.

9. *Exploring Emerging Technologies in the HPC Co-design Space*, Workshop on Accelerator Programming using Directives (WACCPD), New Orleans, 2014.

10. *Exploring Emerging Technologies in the HPC Co-design Space*, FRCRC High Performance Computing Symposium, Laramie, WY, 2013.

11. *On the Road to Exascale: Lessons from Contemporary Scalable GPU Systems*, ATIP/A*CRC/NSF Workshop on Accelerator Technologies for High-Performance Computing, Singapore, 2012.

12. *Emerging Architectures on the Pathway to Exascale Computing*, HP-Cast, Hamburg, 2011.

5.2 Invited

1. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , Queens University, Belfast, 2018.

2. *Working toward performance portability for FPGAs in High Performance Computing*, 3rd International Workshop on FPGA for HPC (IWFH), Tokyo, 2018.

3. *DRAGON: Efficiently Increasing GPU Memory Capacity with NVM* , 5th DOE/MEXT Workshop, Tokyo, 2018.

4. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , National Supercomputing Center in Guangzhou, Guangzhou, 2017.

5. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , ICM, Warsaw, 2017.

6. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , 2017 High Performance Computing (HPC) and Data Analytics Workshop, Hanover, MD, 2017.

7. *Performance Portability for Extreme Scale High Performance Computing* , National Supercomputing Center in Guangzhou, Guangzhou, 2016.

8. *Performance Portability for Extreme Scale High Performance Computing* , International Workshop on HPC Architecture, Software, and Application at an Extreme Scale @ National Supercomputing Center in Wuxi, Wuxi, 2016.

9. *Exploring Emerging Technologies for Extreme Scale High Performance Computing*, École Polytechnique Fédérale de Lausanne (EPFL), Geneva, 2016.

10. *Understanding Portability of a High-Level Programming Model on Diverse HPC Architectures*, ADAC Workshop Switzerland 2016, Lugano, 2016.

11. *Using Application-Specific Performance Models to Inform Dynamic Scheduling*, 6th International Workshop on Runtime and Operating Systems for Supercomputers (ROSS), Kyoto, 2016.

12. *Using Application-Specific Performance Models to Inform Dynamic Scheduling*, 11th Scheduling for Large Scale Systems Workshop, Nashville, 2016.

13. *Managing the Memory Hierarchy*, SOS20, Asheville, 2016.

14. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing*, Forschungszentrum Julich GmbH, Julich Supercomputing Centre (JSC), Julich, 2016.

15. *Managing the Emerging Complexity Wall of Extreme Scale High Performance Computing*, University of Tennessee, Knoxville, Knoxville, 2016.

16. *Preparing for Supercomputing's Sixth Wave*, CHPC National Meeting 2016, East London, South Africa, 2016.

17. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , CHPC National Meeting 2016, East London, South Africa, 2016.

18. *Performance Portability for Extreme Scale High Performance Computing* , Workshop on Clusters, Clouds, and Data for Scientific Computing, Lyon, 2016.

19. *Performance Portability for Extreme Scale High Performance Reconfigurable Computing* , Workshop on FPGAs for Scientific Simulation and Data Analytics, NCSA, 2016.

20. *Exploring Emerging Technologies in the HPC Co-Design Space*, SPPEXA Symposium, Munich, 2016.

21. *Exploring Emerging Technologies in the HPC Co-Design Space*, Barcelona Supercomputing Center, Barcelona, 2015.

22. *Exploring Emerging Technologies in the HPC Co-Design Space*, ISC Workshop on Performance Modeling: Methods and Applications, Frankfurt, 2015.

23. *Quantitatively Modeling Application Resilience with the Data Vulnerability Factor*, Schloss Dagstuhl, Wadern, 2015.

24. *Quantitatively Modeling Application Resilience with the Data Vulnerability Factor*, Salishan Conference on High-speed Computing, Gleneden Beach, OR, 2015.

25. *Exploring Emerging Technologies in the HPC Co-Design Space*, Clusters, Clouds, and Data for Scientific Computing (CCDSC), Lyon, 2014.

26. *Exploring Emerging Technologies in the HPC Co-Design Space*, Institute of Computational Science, Universita della Svizzera italina, Lugano, 2014.

27. *Exploring Emerging Technologies in the HPC Co-design Space*, IBM TJ Watson Research Center, Yorktown, 2014.

28. *Exploring Emerging Technologies in the HPC Co-design Space*, Qatar Computing Research Institute, Doha, 2014.

29. *Exploring Emerging Technologies in the HPC Co-Design Space*, American Physical Society (APS) March Meeting, Denver, 2014.

30. *Exploring Emerging Technologies in the HPC Co-Design Space*, Tokyo Institute of Technology, Tokyo, 2014.

31. *Exploring Emerging Technologies in the HPC Co-Design Space*, JST/CREST International Symposium on Post Petascale System Software, Kobe, 2014.

32. *Exploring Emerging Technologies in the HPC Co-design Space*, 10th International Conf Parallel Processing and Applied Mathematics, Warsaw, 2013.

33. *Exploring Emerging Technologies in the HPC Co-Design Space*, Swiss Federal Supercomputing Centre, Lugano, 2013.

34. *Automated Exploration of the HPC Codesign Space*, International HPC Forum (IHPCF), Changsha, China, 2013.

35. *Intersection of Big Computing and Big Data*, SOS17, Jekyll Island, Georgia, 2013.

36. *Automated Exploration of the HPC Codesign Space with Aspen*, 1st International Workshop on Strategic Development of High Performance Computers, Tsukuba, Japan, 2013.

37. *Creating Effective Teams*, DOE CMD-IT National Laboratories Professional Development Workshop for Underrepresented Participants, ORNL, Oak Ridge, 2013.

38. *Classifying Soft Error Vulnerabilities in Extreme-Scale Scientific Applications Using a Binary Instrumentation Tool*, SIAM CSE Conference, Boston, 2013.

39. *Exploring Emerging Technologies in the HPC Co-Design Space*, NVIDIA GPU Technology Theatre @ SC13, Denver, 2013.

40. *Exploring Emerging Technologies in the HPC Co-Design Space*, Tokyo Institute of Technology Exhibit @ SC13, Denver, 2013.

41. *Exploring Emerging Technologies in the HPC Co-Design Space*, Department of Energy Exhibit @ SC13, Denver, 2013. [slides]

42. *Aspen: a performance modeling language for codesigning Exascale applications and architectures*, Clusters, Clouds, and Data for Scientific Computing, Lyon, 2012.

43. *Emerging Technologies on the Pathway to Exascale*, Institute for Defense Analysis, Bowie, 2012.

44. *SHOC and AWE! An update on the Scalable Heterogeneous Computing Benchmark Suite*, Salishan Conference on High-Speed Computing (Random Access session), Lincoln City, OR, 2012.

45. *Emerging Technologies on the Pathway to Exascale*, PRACE Future Technologies Workshop, Daresbury, 2012.

46. *The Critical Role of Programming Systems in Future Exascale Systems*, CSE Seminar, University of Michigan, Ann Arbor, 2012.

47. *Blackcomb Hardware software Co-design for Nonvolatile Memory in Exascale Systems*, SOS16, Santa Barbara, 2012.

48. *The Critical Role of Software Infrastructure in Future Exascale Systems*, 40th SPEEDUP Workshop, Basel, Switzerland, 2012.

49. *Vancouver – Unlocking the Potential of Scalable Heterogeneous Systems*, DOE Exascale Research Conference, Arlington, 2012.

50. *NSF Keeneland Project*, NCAR Workshop on Programming weather, climate, and earth-system models on heterogeneous multi-core platforms, Boulder, 2011.

51. *Emerging Architectures on the Pathway to Exascale Computing*, Institute for Nuclear Theory: Extreme Computing and its Applications, Seattle, 2011.

52. *The Under-appreciated Exascale Challenge: Memory and Storage*, International Supercomputing Conference, Hamburg, 2011.

53. *Emerging Technologies on the Pathway to Exascale Computing*, USC-DOE conference on Materials for Energy Applications - Experiment, Modeling and Simulations, Los Angeles, 2011.

54. *Programming Heterogeneous Systems Productively*, SOS15, Engelberg, Switzerland, 2011.

55. *Transformational Computational Science: Past, Present, and Future*, Convocation for Georgia Tech School of Computational Science and Engineering, Atlanta, 2011.

56. *The Critical Role of Software Infrastructure in Future Exascale Systems*, NVIDIA GTC Asia, Beijing, 2011.

57. *On the path to Exascale: Deploying an Emerging HPC Architecture*, DOE Exascale Tools Workshop, Annapolis, 2011.

58. *Toward Exascale Computational Science: the Role of Emerging Technologies*, SC10 Broader Engagement Workshop, New Orleans, 2010.

59. *Toward Exascale Computational Science with Graphics Processors*, NVIDIA GPU Technical Conference, San Jose, 2009.

60. *Application Accelerators: Deus ex machina?*, High Performance Embedded Computing Workshop, Lexington, MA, 2009.

61. *Pioneering Science Applications and Breakthroughs on Jaguar*, International Supercomputing Conference, Hamburg, Germany, 2009.

62. *HPC Interconnection Networks*, SOS13, Hilton Head, 2009.

63. *Toward Exascale Computational Science with Graphics Processors*, Tokyo Institute of Technology, Tokyo, 2009.

64. *Toward Exascale Computational Science with Graphics Processors*, Cray Asia Technical Workshop, Tokyo, 2009.

65. *HPC Interconnection Networks – The Key to Exascale Computing*, HPC 2008 – Advanced Research workshop on High Performance Computing, Cetraro, Italy, 2008.

66. *Tightly-coupled heterogeneous computing – deux ex machina*, DOE High Performance Computer Science Week, Denver, 2008.

67. *Performance Modeling and Analysis on Multi-Paradigm Computing Systems Challenges and Opportunities*, Dagstuhl Seminar 07341: Code Instrumentation and Modeling for Parallel Performance Analysis, Wadern, 2007.

68. *The Future of Performance Engineering in HPC*, DOE Advanced Scientific Computing Advisory Committee (ASCAC), Washington, DC, 2007.

69. *Application Accelerators: dues ex machina?*, Purdue University – Computing Research Institute, West Lafayette, 2007.

70. *Application Accelerators: dues ex machina?*, Clusters and Computational Grids for Scientific Computing, Asheville, 2006.

71. *Symbolic Modeling Techniques for Symbolic Modeling Techniques for Next Next-Generation Applications and Generation Applications and Architectures*, DARPA High Productivity Computing Systems Workshop, Denver, 2006.

72. *Overview of Future Technologies at ORNL*, Atomic Weapons Establishment (AWE) seminar, Aldermaston, 2005.

73. *Embracing Uncertainty: Performance analysis and prediction on next-generation computer architectures*, DOE SciDAC Conference, San Francisco, 2005.

74. *Performance Modeling and Analysis: Striking a Balance between Resolution and Insight*, Intel Clustering Workshop, Toronto, 2004.

75. *A Progress Report on the Cray X1 Evaluation at the Center for Computational Sciences at ORNL*, Cray Users Group (CUG) Meeting, Knoxville, TN, 2004.

76. *Performance Analysis of Ultrascale Systems Using Performance Counters*, SIAM Conf. Parallel Processing for Scientific Computing, San Francisco, 2004.

77. *Performance Analysis and Modeling of High End Computing Systems*, Oak Ridge National Laboratory, Oak Ridge, 2003.

78. *Performance Analysis and Modeling of High End Computing Systems*, University of Tennessee Computer Science Department, Knoxville, TN, 2003.

79. *Performance Analysis and Modeling of High End Computing Systems*, HPC User Forum, Princeton, 2003.

80. *Performance Analysis and Modeling of High End Computing Systems*, University of Rochester Computer Science Colloquium, Rochester, NY, 2003.

81. *Performance Analysis and Modeling of High End Computing Systems*, ParCo Symposium on Performance Analysis and Modeling, Dresden, Germany, 2003.

82. *Performance Analysis and Modeling of High End Computing Systems*, Zentrum für Hochleistungsrechnen, Technische Universität Dresden, Dresden, Germany, 2003.

83. *Performance Analysis and Modeling of High End Computing Systems*, Performance Evaluation Research Center (PERC) Workshop on Benchmarking, Performance Analysis and Modeling, Berkeley, CA, 2003.

84. *Performance Analysis and Modeling of Terascale Applications: Striking a Balance between Resolution and Insight*, NSF NPACI All Hands Meeting, San Diego, 2003.

85. *ASCI Experiences: ASCI Tools and Opportunities to Leverage DOE Infrastructure*, NSF NPACI All Hands Meeting, San Diego, 2003.

86. *Performance modeling and analysis: striking a balance between resolution and insight*, SC2003 APART Workshop, Phoenix, 2003.

87. *Scalable performance analysis and prediction techniques for terascale computing*, University of Texas, Austin, Austin, 2002.

88. *Characterization and performance modeling of scalable scientific applications*, Lawrence Berkeley National Laboratory, Berkeley, CA, 2002.

89. *Scalable performance analysis and prediction techniques for terascale computing*, DARPA HPCS PI Meeting, Phoenix, 2002.

90. *Automated techniques for terascale performance analysis*, Lawrence Berkeley National Laboratory, Berkeley, CA, 2001.

91. *A performance data explosion: striking an appropriate balance between resolution and insight*, LACSI Performance Tools Workshop, Santa Fe, 2001.

92. *Techniques for high performance computational steering*, Microsoft Research, Seattle, 1999.

93. *Techniques for high performance computational steering*, University of New Mexico, Albuquerque, 1999.

94. *Techniques for high performance computational steering*, Lawrence Livermore National Laboratory, Berkeley, CA, 1999.

95. *Techniques for high performance computational steering*, University of Arizona, Tucson, 1999.

96. *Techniques for high performance computational steering*, University of Minnesota, Twin Cities, 1999.

5.3 Contributed

1. *CLACC: OpenACC for LLVM*, EuroLLVM, Bristol, 2018.

2. *Performance Engineering of Emerging Memory Systems* , SIAM PP18 MS41, Tokyo, 2018.

3. *Exploring Extreme Heterogeneity with Analytical Modeling*, SIAM PP18 MS111, Tokyo, 2018.

4. *Programming Nonvolatile Memory Systems*, Salishan Conference on High Speed Computing (Random Access Talk), Gleneden Beach, Oregon, 2016.

5. *Understanding Portability of a High-Level Programming Model on Diverse HPC Architectures*, DOE Centers of Excellence Performance Portability Meeting, Glendale, AZ, 2016.

6. *Exploring Emerging Technologies in the HPC Co-Design Space*, Exascale Application and Software Conference, Edinburgh, 2015. (Best Presentation Award).

5.4 Panels

1. *Computing in the Post-Moore Era*, SIAM PP18, Tokyo, 2018.
2. *Reconf500: Can we join the 500 club?*, 3rd International Workshop on FPGA for HPC (IWFH), Tokyo, 2018.
3. *Post Moore Supercomputing*, SC17, Denver, 2017.
4. *Post Moores Era Supercomputing*, SC16, Salt Lake City, 2016.
5. *Opportunities for HPC based on OpenPOWER Technologies*, International Workshop on OpenPOWER for HPC (IWOPH'16) @ ISC16, Frankfurt, 2016.
6. *Parallel Programming*, ISC, Frankfurt, 2016.
7. *Towards Exascale Memory Systems*, 17th SIAM Conference on Parallel Processing for Scientific Computing, Paris, 2016.
8. *Post Moore's Era Supercomputing in 20 Years (PMES) Panel*, SC16 Panel, Salt Lake City, 2016.
9. *HPC and Big Data Convergence*, 2nd International HPC Forum, Tianjin, 2015.
10. *Exascaling your Applications*, NVIDIA GTC 2012, San Jose, 2012.
11. *Working with Funding Agencies*, NSF Academic Workshop in Computing for Underrepresented Participants, Atlanta, 2012.
12. *DARPA UHPC: Echelon and CHASM*, HPCC, Newport, RI, 2011.
13. *Concurrency-Related Challenges*, Advanced Computing Systems Research Program, Annapolis, 2009.
14. *Programming Accelerator Clusters — Where Do We Go From Here?*, Workshop on Parallel Programming on Accelerator Clusters (PPAC), New Orleans, 2009.
15. *A Multidisciplinary Approach to Resilience*, International Supercomputing Conference, Hamburg, Germany, 2009.
16. *Is PGAS right for heterogeneous computing*, PGAS Conference, Ashburn, VA, 2009.
17. *Past, Present, and Future of Parallel Processing – ICPP 35th Anniversary Panel*, International Conf. Parallel Processing, Columbus, 2006.
18. *How video games and multimedia will affect the HPC industry?*, International Conf. Parallel Processing, Oslo, 2005.
19. *Future Directions for HPC*, SOS9 Workshop, Davos, 2005.
20. *High Performance Computing: Successes, Failures, and Future Directions*, ACS/IEEE International Conference on Computer Systems and Applications, Cairo, 2005.
21. *OS and System Software for Ultrascale Architectures*, SOS8 Workshop, Charleston, 2004.
22. *A Science-Based Path to Petascale Computing*, Intel HPC Roundtable, Portland, OR, 2004.
23. *HPCchallenge Benchmarks: An Expanded View of High End Computer (HEC) Performance*, SC2004 High Performance Computing, Networking, and Storage, Pittsburgh, 2004.
24. *Design and evaluation of future computing architectures*, Fall Creek Falls Computing workshop, Fall Creek, TN, 2004.

6 Software Products

1. DESTINY: a 3D design-space exploration tool for SRAM, eDRAM and non-volatile memory. https://code.ornl.gov/3d_cache_modeling_tool/destiny
2. Open Accelerator Research Compiler (OpenARC). http://ft.ornl.gov/research/openarc
3. SHOC – Scalable HeterOgeneous Computing Benchmark Suite. https://github.com/vetter/shoc
4. mpiP – Lightweight, scalable MPI profiling tool. Verified to over 200k processors. http://www.llnl.gov/casc/mpip
5. Sequoia – Tool for application performance modeling and characterization.
6. Umpire – Tool for verifying the correctness of MPI applications dynamically at runtime. Inspired Intel's Message Checker™ in the Intel Cluster Toolkit™.
7. Photon – Novel implementation of MPI that supports statistical message sampling for performance analysis.
8. Performance assertions – Annotation and runtime toolkit for explicit assertion of performance expectations in HEC applications.

7 Patents

1. *Automatic classification of communication inefficiencies in distributed applications.* US Patent 6,850,920. Application awarded February 2005.

8 Professional Activities and Service

8.1 Leadership

1. Third International Workshop on Post Moores Supercomputing, Co-chair, 2018.
2. ASCR Workshop on Extreme Heterogeneity in High Performance Computing, Chair, 2018.
3. HPC Asia 2018: International Conference on High Performance Computing in Asia Pacific Region, Steering Committee, 2018.
4. Second International Workshop on Post Moores Supercomputing, Co-chair, 2017.
5. SC17, Birds-of-a-Feather Chair, 2017.
6. Dagstuhl Seminar 17431: Performance Portability in Extreme Scale Computing: Metrics, Challenges, Solutions, Co-organizer, 2017.
7. Workshop on HPC Computing in a Post Moore's Law World (HCPM), Co-chair, 2017.
8. ISC, Research Posters, Chair, 2017.
9. IEEE Micro: Special Issue on Architectures for the Post-Moore Era, Co-Guest Editor, 2017.
10. ACM Journal on Emerging Technologies in Computing (JETC), Associate Editor, 2016.
11. International Supercomputing Conference (ISC), Steering Committee, 2016.
12. Exascale Requirements Review for Advanced Scientific Computing Research (ASCR), Co-chair, 2016.
13. First International Workshop on Post Moores Supercomputing, Co-chair, 2016.

14. SC, Steering Committee, 2015.
15. SC15, Technical Program Chair, 2015.
16. ACM Gordon Bell Prize, Deputy Chair, 2015.
17. Workshop on accelerator programming using directives (WACCPD), Steering Committee, 2014.
18. SC14, Technical Program Deputy Chair, 2014.
19. SC14, Emerging Technologies Co-chair, 2014.
20. U. Tsukuba (Japan), Center for Computational Sciences, Chair of External Review Board, 2014.
21. SC12, Technical Papers Co-chair, 2012.
22. ACM SIGPLAN Workshop on Memory Systems Performance and Correctness (MSPC), General Chair, 2011.
23. MTAAP 2011, Co-chair, 2011.
24. ACM SIGPLAN Workshop on Memory Systems Performance and Correctness, Steering Committee, 2010.
25. Cross-cutting Technologies for Computing at the Exascale Workshop, Programming Models Theme Leader, 2010.
26. IPDPS 2010 - IEEE International Parallel and Distributed Processing Symposium, Vice chair for applications, 2010.
27. MTAAP 2010 - Workshop on Multithreaded Architectures and Applications, Co-chair, 2010.
28. DOE Institute for Advanced Architectures and Algorithms - Interconnections Networks Workshop, Co-chair, 2008.
29. IEEE Transactions on Computers, Associate Editor, 2007.
30. International Conference on Parallel Processing (ICPP), Vice-Chair for Software Systems, 2007.
31. SC07, Disruptive Technologies Chair, 2007.
32. DOE Software Development Tools for Petascale Computing Workshop, General Chair, 2007.
33. Euro-Par 2006, Vice Chair for Performance, 2006.
34. SC05, Technical Papers Area Chair for Performance, Tutorials Committee, 2005.
35. International Conference on Parallel Processing (ICPP), Vice-Chair for Performance Evaluation, 2004.
36. Euro-Par 2004, Vice Chair for Performance Evaluation, 2004.
37. Los Alamos Computer Science Institute (LACSI) Symposium Workshop on Performance Analysis with R. Barrett and A. Malony, Co-chair, 2002.
38. Los Alamos Computer Science Institute (LACSI) Symposium Workshop on Performance Analysis with R. Barrett and A. Malony, Co-chair, 2001.

8.2 Service

1. IEEE Computer Society Fellow Evaluation Committee, Program Committee, 2018.
2. PPOPP, Program Committee, 2019.
3. PPOPP, Program Committee, 2018.
4. SIAM Parallel Processing Conference, Organizing Committee, 2018.
5. MCHPC2017: Workshop on Memory Centric Programming for HPC, Program Committee, 2017.
6. 3rd Workshop on Heterogeneous High-performance Reconfigurable Computing , Program Committee, 2017.
7. High Performance Extreme Computing Conference (HPEC), Program Committee, 2017.
8. Re-Emergence of Vector Architectures (REV-A), Program Committee, 2017.
9. International Symposium on Performance Analysis of Systems and Software (ISPASS), Program Committee, 2017.
10. PPOPP, Program Committee, 2017.
11. International Parallel and Distributed Processing Symposium (IPDPS), Program Committee, 2017.
12. SC16, Program Committee, Technical Papers, 2016.
13. ICS, Program Committee, 2016.
14. ISC, Program Committee, 2016.
15. PMMA16, Organizing Committee, 2016.
16. MODSIM, Organizing Committee, 2016.
17. High Performance Extreme Computing Conference, Program Committee, 2016.
18. MODSIM, Organizing Committee, 2015.
19. ASPLOS, Program Committee, 2015.
20. High Performance Extreme Computing Conference, Program Committee, 2015.
21. DOE ASCAC Workforce Subcommittee, Advisory/Review Committee, 2014.
22. PPOPP, Program Committee, 2014.
23. ICS, Program Committee, 2014.
24. DOE MODSIM 2014, Organizing Committee, 2014.
25. MTAAP, Program Committee, 2014.
26. High Performance Extreme Computing Conference, Program Committee, 2014.
27. MTAAP, Program Committee, 2013.
28. International Parallel and Distributed Processing Symposium (IPDPS), Program Committee, 2013.
29. GPGPU6, Program Committee, 2013.
30. SC13, Panels Committee; Gordon Bell Committee; Emerging Technologies Committee, 2013.
31. HiPC, Program Committee, 2013.

32. DOE MODSIM 2013, Organizing Committee, 2013.
33. MSPC 2013, ACM SIGPLAN Workshop on Memory Systems Performance and Correctness, Program Committee, 2013.
34. High Performance Extreme Computing Conference, Program Committee, 2013.
35. ISC 2012, Program Committee, Tutorials, 2012.
36. ACM ICS, Program Committee, 2012.
37. ACM InPar, Program Committee, 2012.
38. PPOPP, Program Committee, 2012.
39. ACM Workshop on General Purpose Processing on Graphics Processing Units, Program Committee, 2011.
40. SC11, Program Committee, Tutorials, 2011.
41. SciDAC 2011, Organizing Committee, 2011.
42. ACM Workshop on Emerging Applications and Many-Core Architectures (EAMA), Organizing Committee, 2011.
43. First International Workshop on Characterizing Applications for Heterogeneous Exascale Systems, Program Committee, 2011.
44. SC10, Program Committee, Tech Papers, 2010.
45. SIAM Parallel Processing Conference, Organizing Committee, 2010.
46. Workshop on Parallel Programming on Accelerator Clusters (PPAC), Program Committee, 2009.
47. SC09, Program Committee, Tech Papers, 2009.
48. DOE Institute for Advanced Architectures and Algorithms - Workshop on HPC Architectural Simulation (HPCAS-1), Organizing Committee, 2009.
49. MTAAP, Program Committee, 2009.
50. International Exascale Software Project, Organizing Committee, 2008.
51. SC08, Tutorials Committee, Disruptive Technology Committee, 2008.
52. SIAM Parallel Processing Conference, Mini-symposium Organizer, 2008.
53. IEEE International Symposium on Workload Characterizations, Program Committee, 2008.
54. Workshop on Multithreaded Architectures and Applications (MTAAP), Program Committee, 2008.
55. Principles and Practice of Parallel Programming (PPOPP), Program Committee, 2008.
56. ACM Computing Frontiers, Program Committee, 2007.
57. IEEE International Symposium on Workload Characterizations, Program Committee, 2007.
58. Institute for Advanced Architectures Workshop on Memory Opportunities for High Performance Computing, Program Committee, 2007.
59. DOE/DOD Workshop on Emerging High Performance Architectures and Applications, Organizing Committee, 2007.
60. SC06, Exotic Technologies Initiative Committee, Technical Papers Committee, 2006.

61. International Parallel and Distributed Processing Symposium (IPDPS), Program Committee, 2006.
62. Parallel Computing (ParCo), Program Committee, 2005.
63. ACM SIGMETRICS, Program Committee, 2005.
64. High Performance Distributed Computing (HPDC), Program Committee, 2005.
65. LCI International Conference on Linux Clusters: The HPC Revolution, Program Committee, 2005.
66. Workshop on System Management Tools for Large-Scale Parallel Systems, Program Committee, 2005.
67. International Parallel and Distributed Processing Symposium (IPDPS), Program Committee, 2005.
68. Seventh Workshop on Languages, Compilers, and Runtime Support for Scalable Systems (LCR), Program Committee, 2004.
69. International Conference for High-Performance Computing, Program Committee, 2004.
70. International Parallel and Distributed Processing Symposium (IPDPS), Program Committee, 2003.
71. International Conference on Computational Science, Workshop on Performance Evaluation, Modeling and Analysis of Scientific Applications on Large-Scale Systems, Program Committee, 2003.
72. SC03, Program Committee, 2003.
73. Parallel Computing (ParCo), Program Committee, 2003.
74. Los Alamos Computer Science Institute Symp, Program Committee, 2003.
75. National Science Foundation, Proposal Review Panelist, 2003.
76. Department of Energy. Advanced Scientific Computing Research (ASCR) proposals, Proposal Review Panelist, 2002.
77. LCI International Conference on Linux Clusters: The HPC Revolution, Program Committee, 2002.
78. Department of Energy NERSC, Computational Review Panel, 2002.
79. Georgia Tech Center for Experimental Research in Computer Systems, Advisory Panel, 2002.
80. National Science Foundation, Proposal Review Panelist, 2001.
81. Science Foundation Ireland, Proposal Review Panelist, 2001.
82. Lawrence Livermore National Laboratory University Collaborative Research Program proposals, Proposal Review Panelist, 2000.
83. National Science Foundation, Proposal Review Panelist, 1998.

9 Teaching and Student Guidance

9.1 Teaching

1. Fall 2007, Georgia Tech CS6230, Graduate-level Introduction to HPC Tools and Applications.
2. Fall 2010, Georgia Tech CS6230, Graduate-level Introduction to HPC Tools and Applications.
3. Spring 2012, Georgia Tech CSE6220, Graduate-level Introduction to HPC.

9.2 Postdoctoral Fellows Supervised

1. Dr. Sadaf Alam, Oak Ridge National Laboratory
2. Dr. Anthony Danalis, Oak Ridge National Laboratory and University of Tennessee
3. Dr. Mehmet Beliveranli, Oak Ridge National Laboratory, 2016–present
4. Dr. Jungwon Kim, Oak Ridge National Laboratory
5. Dr. Graham Lopez, Oak Ridge National Laboratory
6. Dr. Collin McCurdy, Oak Ridge National Laboratory
7. Dr. Sparsh Mittal, Oak Ridge National Laboratory, 2014–2016
8. Dr. Ivy "Bo" Peng, Oak Ridge National Laboratory, 2017–present
9. Dr. Kittisak Sajjapongse, Oak Ridge National Laboratory, 2016
10. Dr. Srinivas Sridharan, Oak Ridge National Laboratory

9.3 M.S. Committee Member

1. Sundaresan Venkatasubramanian, Georgia Institute of Technology, 2009

9.4 Ph.D. Committee Member

1. Srinivas Sridharan, University of Notre Dame
2. Wenbin Zhu, University of New Mexico, 2008

9.5 M.S. Students Supervised

1. Viraj Rameshrao Paropkari, Georgia Institute of Technology, 2007
2. Kyle Spafford, Georgia Institute of Technology, 2008-9

9.6 Student Internships Supervised

1. Sreesa Akella, 2005, University of South Carolina-Columbia
2. Rshiraj Bheda, 2015, 2016, Georgia Tech
3. Yong Chen, 2010, Illinois Inst. of Technology
4. Hsiang-Yun Cheng, 2014, Penn State University
5. Luis Cordova, 2005, University of South Carolina-Columbia
6. Tristan Cuevas, 2008, Kent State University
7. Anthony Danalis, 2006, University of Delaware
8. Ashley DeFlumere, 2008, Mount Holyoke College
9. Keisuke Fukuda, 2013, 2014, Tokyo Tech
10. Andrew Gearhart, 2008, University of Delaware
11. Seonglyong Gong, 2016, University of Texas, Austin
12. Randy Hassell, 2007, Old Dominion University
13. Tetsuya Hoshino, 2014, Tokyo Tech
14. Matthew Jallouk, 2011, 2012, 2013, University of Tennessee-Chattanooga
15. David LaBissoniere, 2006, East Tennessee State University
16. Joseph Lambert, 2006, Bucknell University

17. Matthew Larsen, 2014, University of Oregon
18. Robert Lim, 2014, University of Oregon
19. Charles Lively, 2008, Texas A&M University
20. Kelly Livingston, 2008, University of Delaware
21. Nicholas Murphy, 2007, University of Washington
22. Viraj Paropkari, 2007, Georgia Institute of Technology
23. David Poliakoff, 2012, University of Oregon
24. Matthew Poremba, 2014, Penn State University
25. Amit Sabne, 2014, Purdue University
26. Putt Sakdhnagool, 2014, Purdue University
27. Samuel Schlegel, 2015, Florida Atlantic University
28. Kyle Spafford, 2008, Georgia Institute of Technology
29. Srinivas Sridharan, 2008, University of Notre Dame
30. Junqing Sun, 2007-8, University of Tennessee-Knoxville
31. Philip Taffet, 2014, Rice University
32. Nirmal Thacker, 2008, Georgia Institute of Technology
33. Mariam Umar, 2016, Virginia Tech
34. Karthik Vijayakumar, 2009, North Carolina State University
35. Rujia Wang, 2016, University of Pittsburgh
36. Bo Wu, 2013, College of William and Mary
37. Panruo Wu, 2013, University of California – Riverside
38. Jake Wynne, 2015, University of Tennessee
39. Li Yu, 2014, Illinois Institute of Technology

10 Funding

10.1 Current

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2017	1	DOE	Preparing for Extreme Heterogeneity in High Performance Computing [Workshop]	$40	OC Member
2017	3	DOE	Sawtooth: Investigating solutions to memory and storage challenges	$12000	PI
2017	3	DOE	RAPIDS: SciDAC4 Computer Science and Data Institute	$18000	Senior Personnel
2017	3	DOE	Methods and Interfaces for Quantum Acceleration of Scientific Applications	$4500	Senior Personnel
2017	3	DOE	Heterogeneous Quantum Dynamics Simulator	$4500	Senior Personnel

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2017	3	DOE	Panorama360: Performance data capture and analysis for end-to-end scientific workflows	$3000	Co-PI
2016	2	LDRD	Tahoe: Designing and programming exascale memory hierarchies	$720	PI
2016	3	DOE	Exascale Computing Project	$n/a	Manager
2016	3	DOE	PROTEAS: Programming Systems for Emerging Architectures	$4500	PI
2014	3	DOE	Panorama: Predictive Modeling and Diagnostic Monitoring of Extreme Science Workflows	$4500	Co-PI
2014	3	DOE	ARES: Abstract Representations for Extreme-Scale Software Stack	$3000	Co-PI
2013	3	DOE	Oxbow: Synthesizing architectural features for extreme-scale co-design applications	$2250	PI
2013	3	DOE	Uncertainty Quantification Methodologies for Enabling Extreme-Scale Science	$2250	Co-PI

10.2 Pending

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role

10.3 Completed

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2013	3	DOE	Blackcomb2: Hardware-Software Co-design for Non-Volatile Memory in Exascale System	$2100	PI
2013	3	DOE	Vancouver2: Designing a Next-Generation Software Infrastructure for Productive Heterogeneous Exascale Computing	$2100	PI
2011	5	DOE	Center for Exascale Simulation of Advanced Reactors (CESAR) Co-Design Center	$20000	Co-PI
2011	5	DOE	Exascale Co-Design Center for Materials in Extreme Environments (ExMatEx)	$20000	Co-PI
2010	3	DOE	Blackcomb: Hardware-Software Co-design for Non-Volatile Memory in Exascale System	$3000	PI
2010	3	DOE	Vancouver: Designing a Next-Generation Software Infrastructure for Productive Heterogeneous Exascale Computing	$2600	PI
2010	6	NVIDIA	NVIDIA CUDA Center of Excellence (industry grant)	$700	PI
2010	1	Reservoir Labs	Advance Static and Dynamic Mapping of HPC Applications on Petascale Systems	$60	Co-PI
2010	2	DARPA	Echelon: Extreme-scale Compute Hierarchies with Efficient Locality-Optimized Nodes	$25000	Co-PI
2010	2	DARPA	CHASM: Challenge Applications and Scalable Metrics for Ubiquitous High Performance Computing	$7500	Co-PI
2010	1.5	DOE	SciDAC-e: Enhancing Productivity of Materials Discovery Computations for Solar Fuels and Next Generation Photovoltaics	$110	Co-PI
2009	5	NSF	Keeneland: National Institute for Experimental Computing / 0910735 / Vetter	$12000	PI
2009	4	NSF	CI-ADDO-NEW: MANIFOLD- A Scalable Simulation Infrastructure for Future Many Core Systems	$1300	
2009	3	DOE	MOGO: Model Oriented Global Optimization of Petascale Scientific Applications	$1500	PI

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2008	3	DARPA	Blackjack: Evaluation of Architecture-Aware Compiler Environments	$4500	Co-PI
2008	4	DARPA	Productivity Evaluation of HPCS Architectures	$3600	PI
2008	1	Advanced Micro Devices	Geneva: Performance Prediction for Tightly Coupled Heterogeneous Processors	$50	
2007	3	NSF	NSF CRI: Development of a Research Infrastructure for the Multithreaded Computing Community Using the Cray Eldorado Platform	$80	
2006	2	ORNL LDRD	An Evolutionary Framework for Porting Applications to Petascale Platforms	$500	
2006	5	DOE	SciDAC Performance Engineering Research Institute	$15000	Co-PI
2006	5	DOE	SciDAC Scientific Data Management Center	$16500	Senior Personnel
2005	2	ORNL LDRD	Automated Performance Tools for Petascale Systems with Lightweight Compute Node Kernels	$550	
2004	4	National Security Agency	Petascale Single System Image	$1500	
2004	3	DOE	Modular Linux and Adaptive Runtime support for HEC OS/R Research	$1500	
2004	2	ORNL LDRD	Exploring Alternative Technologies for Next-Generation Leadership Class Computing	$	
2004	4	DOE	HPCS PetaScale Execution Time Evaluation	$900	
2004	3	DOD HPC-MOD	Specialized Performance Analysis and Modeling of Department of Defense High Performance Computing Modernization Program Applications	$540	
2004	2	DOE	Performance Evaluation Research Center	$	

10.4 Not funded

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2018	1	NSF	Collaborative: SI2-S2I2 Conceptualization: Software Institute for Robust Science: Reproducibility, Automation and Trust in Software for Science	$	Co-PI
2017	3	DOE	Quantum Sampling Algorithms for Scientific Applications of Machine Learning	$4500	Senior Personnel
2016	3	DOE	Automation for Robust Exascale Workflow Applications	$3750	Co-PI
2016	3	DOE	Montreux: Performance Characterization, Prediction, and Measurement of ECP Applications	$2250	PI
2015	4	DARPA	CATALYST (Building Resource Adaptive Software Systems for Heterogeneous, Scalable Computing Systems)	$8500	PI
2015	3	DOE	Ashland: Globally Optimized Dynamic Resource Management for Distributed Scientific Workflows	$4500	PI
2015	3	DOE	Silverburn: Measuring, Understanding, and Predicting Behaviors of Emerging Extreme-scale Storage Paradigms	$2420	PI
2015	3	DOE	Whistler: Designing Resilient Next-Generation Memory Systems	$2100	Co-PI
2015	3	DOE	Resilience Building Blocks	$3750	Co-PI
2014	5	NSF	FG Cloud	$10000	Co-PI
1	3	DOE	IDEAS: Interoperable Design of Extreme-scale Application Software	$12000	Senior Personnel

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2013	3	DOE	ExaMover: Adaptive Network Transport and Unified I/O Management Framework for Moving Exabyte Data over WAN	$1800	PI
2013	3	NSF	ABI development: Improving Protein Domain Modeling using Ultrascale Computing	$1100	Co-PI
2013	5	DOE	E3-S: Exascale End-to-End Solvers for PDE-Based simulation and optimization problems	$4500	Co-PI
2012	2	ARO	Pinehurst	$1500	PI
2012	5	DOE	Towards Predictive Exascale Simulations of Materials Interfaces in Extreme Environments	$1350	Co-PI
2011	5	DOE	SciDAC3: IDEAL: Institute for Data movement and Energy Optimization in Applications (IDEAL)	$3000	Co-PI
2011	3	NSF	G8: Tempo	$600	Co-PI
2011	3	NSF	G8: XMG	$600	Co-PI
2010	3	NSF	FastReduce: Infrastructure for end-to-end exascale fast multipole methods	$500	
2010	3	DOE	Whistler: Integrated Interconnection Networks for Exascale Computing	$100	
2010	3	DOE	Breckenridge: GPU Memory Hierarchies for Exascale Systems	$3600	
2010	3	DOE	An Open Integrated Software Stack for Extreme Scale Computing	$15000	
2010	3	DOE	Large Data Systems – Computational Data Flows for Exascale Machines	$2700	
2010	5	DOE	Fusion Reactor Exascale Simulation Co-design Center (FRESCO)	$40000	Co-PI
2010	5	DOE	Combustion Exascale Co-Design Center	$40000	Co-PI
2010	5	DOE	Exascale Center for Electrochemical Energy storage Devices (ExCEED)	$40000	
2010	3	NSF	Camden: Accelerating the deployment of the Keeneland ID system for production availability	$3000	
2010	1.5	DOE	SciDAC-e: Accelerating Simulations of Interfacial Transport and Reactivity in Energy Materials	$700	
2009	3	NSF	A Scalable Metadata Management Framework for ExaByte File Systems	$1000	
2009	3	DOE	Designing a Distributed InfiniBand Network for Petascale Science	$1500	
2009	3	DOE	Exascale Computing of Materials under Extreme Conditions for Energy Technologies	$4500	
2009	3	DOE	Novel Parallel Numerical Algorithms for Emerging Exascale Computing Platforms	$3300	
2009	3	NSF	Heterogeneous Virtual Machine: Future Execution Environments for Heterogeneous Many-core Architectures	$1000	
2009	4	NSF	Manifold – A Scalable Simulation Infrastructure for Future Many Core Systems	$1679	
2009	3	DOE	Extreme-Scale Correctness Tools	$1800	
2008	1.5	DOE	IAA Interconnection Networks for Exascale Computing	$1000	
2007	3	DOE	System Software for Tightly Coupled Heterogeneous Computing	$2070	
2007	3	DOE	Breaking the Collective Wall for Ultra-Scale Parallel IO	$2099	
2007	3	DOE	Global Storage-Class Memory to Enable Efficient PetaByte Data Accesses for PetaScale Scientific Applications	$2000	
2007	2	ORNL LDRD	An Integrated Framework to Improve Productivity and Performance on Emerging Processor Architectures	$700	
2007	2	ORNL LDRD	Enabling Efficient Parallel IO for UltraScale Computing	$800	
2007	3	DOE	High Performance Computing Ecosystem Service Oriented Architecture (HPCE SOA)	$1140	
2006	1.5	NSF	National Innovation Collaboration Ecosystem (NICE) Computing Convergence Study (CCS)	$200	

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2006	5	DOE	Center for Scientific Data-In-Transit	$	
2006	5	DOE	Real-Time Data Reduction with Field Programmable Gate Arrays (FPGAs)	$	
2006	5	DOE	Providing Solutions to Critical Software Gaps to Petascale Science	$	
2006	5	DOE	Institute for Enabling High-Performance Applications on Emerging Architectures	$	
2006	5	DOE	Chemistry SAP	$	

11 Personal

1. Citizenship: USA.
2. Security clearances: Available upon request.

12 References

- See attached reference list.